Propeller De-icing Brush Module Assembly (Mod 6/1674) – Maintenance Practices

1. Servicing

Refer to Figure 1 for details.

A. Replacement of Brush Retainer Assembly (B. F. Goodrich P/N 3E2114–1)

CAUTION **BRUSH MODULES ARE PACKED WITH A PROTECTIVE CARDBOARD COVER AND ELASTIC BAND WHICH RETAINS THE BRUSHES UNDER SPRING TENSION. FAILURE TO RESTRAIN BRUSHES WHEN REMOVING PROTECTIVE COVER OR ALLOWING BRUSHES TO SPRING FORWARD COULD RESULT IN BRUSH DAMAGE.**

1) Remove brush module assembly from engine mounting bracket, refer to Para 2.A.

2) Remove and retain four screws, nuts and washers securing connector to bracket. Note orientation of connector relative to bracket, so that new or replacement retainer assembly connector will be replaced in exact same position.

3) Remove and retain two screws, flat washers, lock washers and nuts securing brush retainer assembly to bracket.

4) Discard brush retainer assembly.

Note Refer to Figure 1 and check that brush modules are stacked in correct sequence as follows:

a) Observe connector pin identification and physically check wiring to determine related module A, B, C.

b) Ensure that modules are stacked in A, B, C sequence with module A positioned against bracket.

5) Secure brush retainer assembly to bracket with two screws, flat washers, lock washers and nuts.

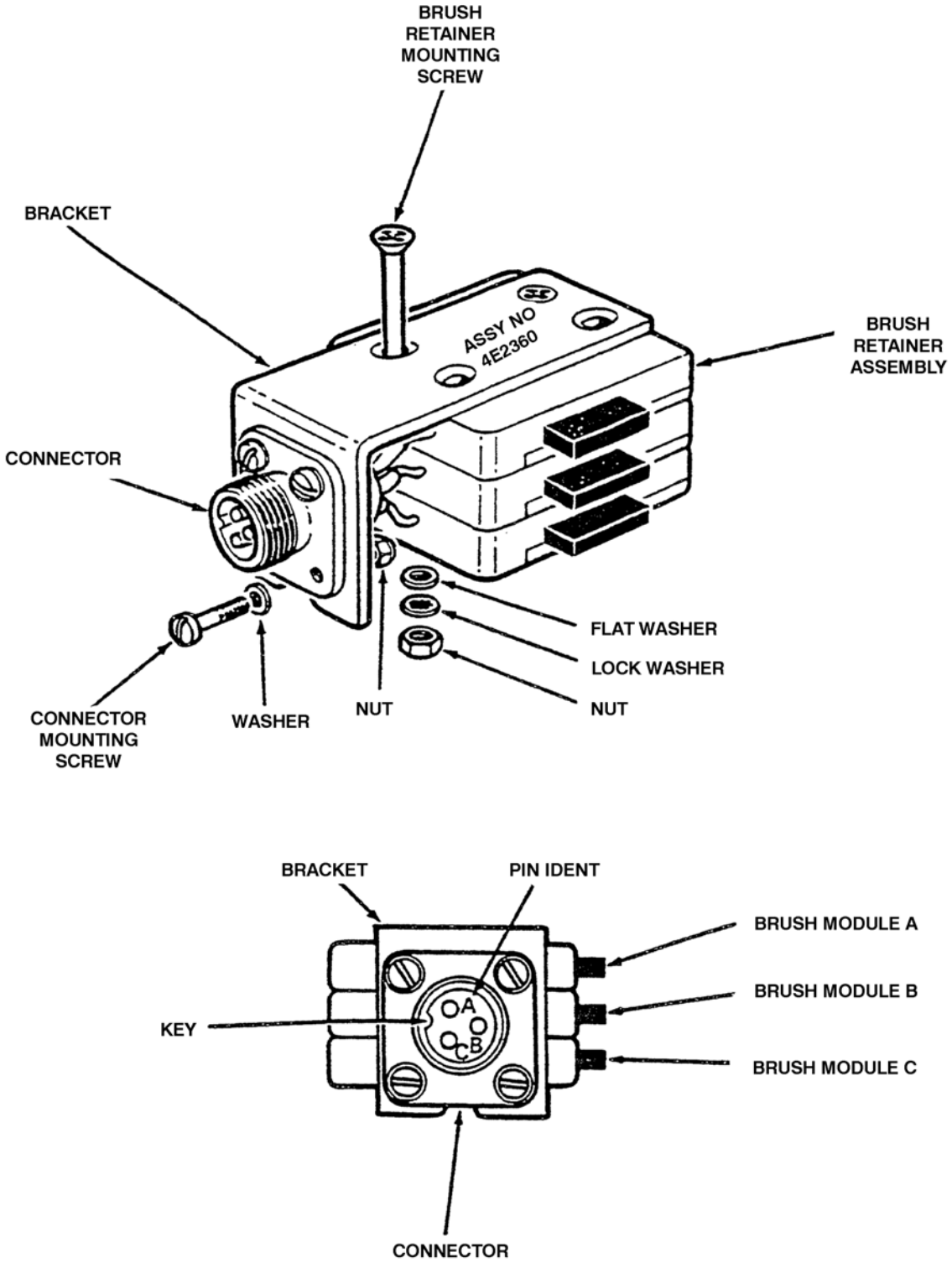
6) Ensure brush modules are square to one another then tighten screws.

7) Check brushes for freedom of movement.

8) Ensure that connector orientation to bracket is as noted in Para 1.A.2).

9) Secure connector to bracket with four screws, washers and nuts.

10) Install brush module assembly on engine mounting bracket, refer to Para 2.B.



Brush Repair Assembly – Removal/Installation
Figure 1

2. Removal/Installation

Refer to Figure 2 for details.

A. Remove Brush Module Assembly

1) Ensure that PROPELLER DEICE circuit breaker is disengaged.

2) Disconnect electrical connector from brush module assembly.

3) Remove and retain two nuts, four washers and two bolts then remove brush module and shim (if shim is installed). Retain shim.

B. Install Brush Module Assembly

CAUTION **FAILURE TO EXERCISE EXTREME CARE WHEN HANDLING, OR IMPOSING SIDE LOADS ON BRUSHES DURING INSTALLATION COULD RESULT IN BRUSH DAMAGE.**

1) Position brush module assembly on engine mounting bracket and insert bolts with washers under bolt heads through module and bracket. Do not install nuts and washers at this time.

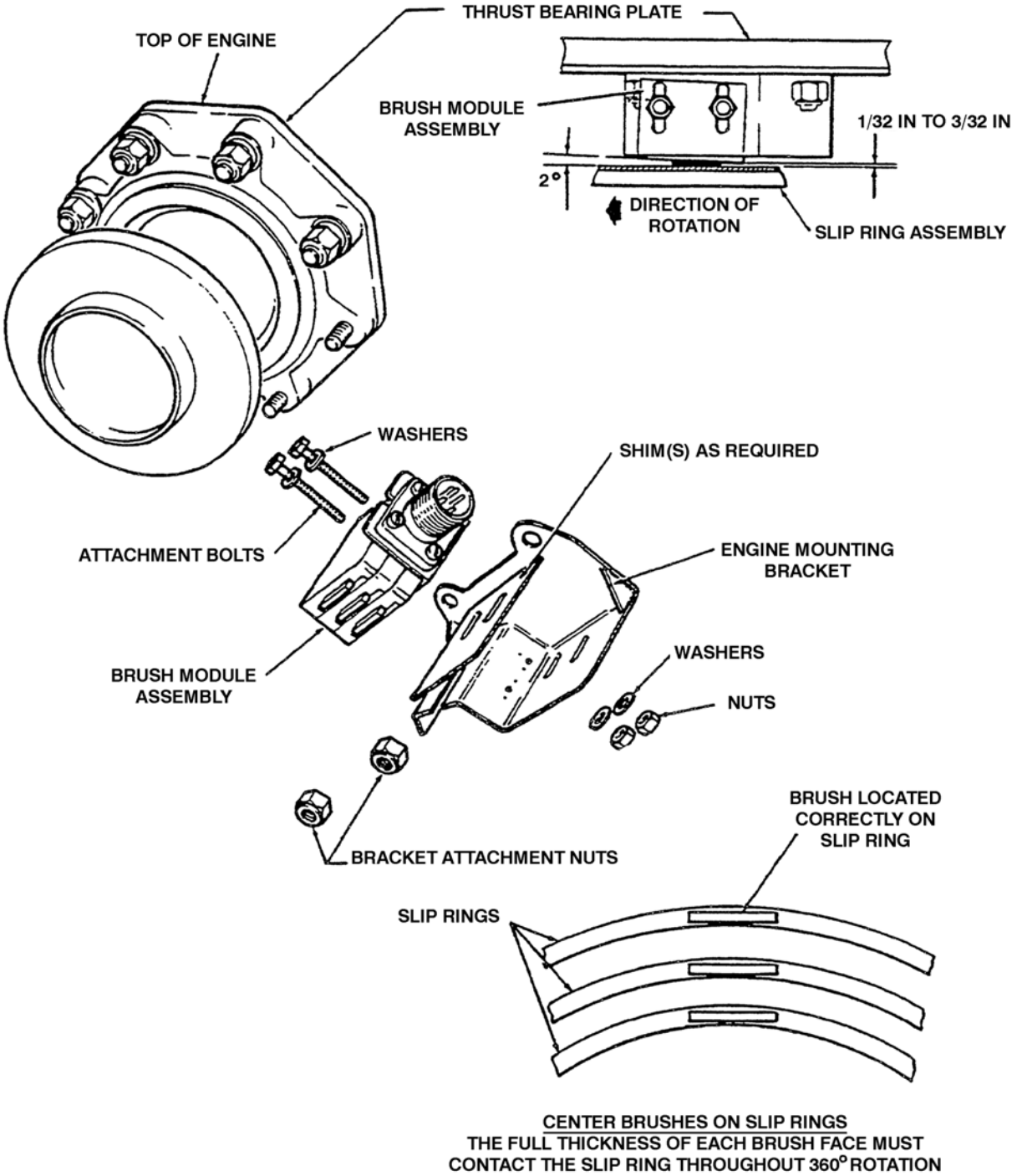
2) Check that brushes are aligned with propeller slip rings so that entire brush face is in contact with copper rings. Check alignment through entire 360 degrees of slip ring rotation. If not aligned install one or more shims as required to align brushes to approximate center of slip rings.

3) Install nuts and washers, and tighten to secure brush module assembly to engine mounting bracket. A minimum distance of 1/32–inch and a maximum of 3/32–inch must be maintained between brush module assembly and slip ring surface.

4) Brush module assemblies must be positioned so that brushes contact slip rings at an angle of approximately 2°, to prevent chattering that could cause brushes to break up (refer to Figure 2). The angle is measured toward the direction of rotation of slip rings.

Note On all installations, whether new or replacement, allow a run-in period of at least one hour before operating the propeller de-icing system.

5) Connect electrical connector to brush module assembly.



Brush Module Assembly – Removal/Installation
Figure 2

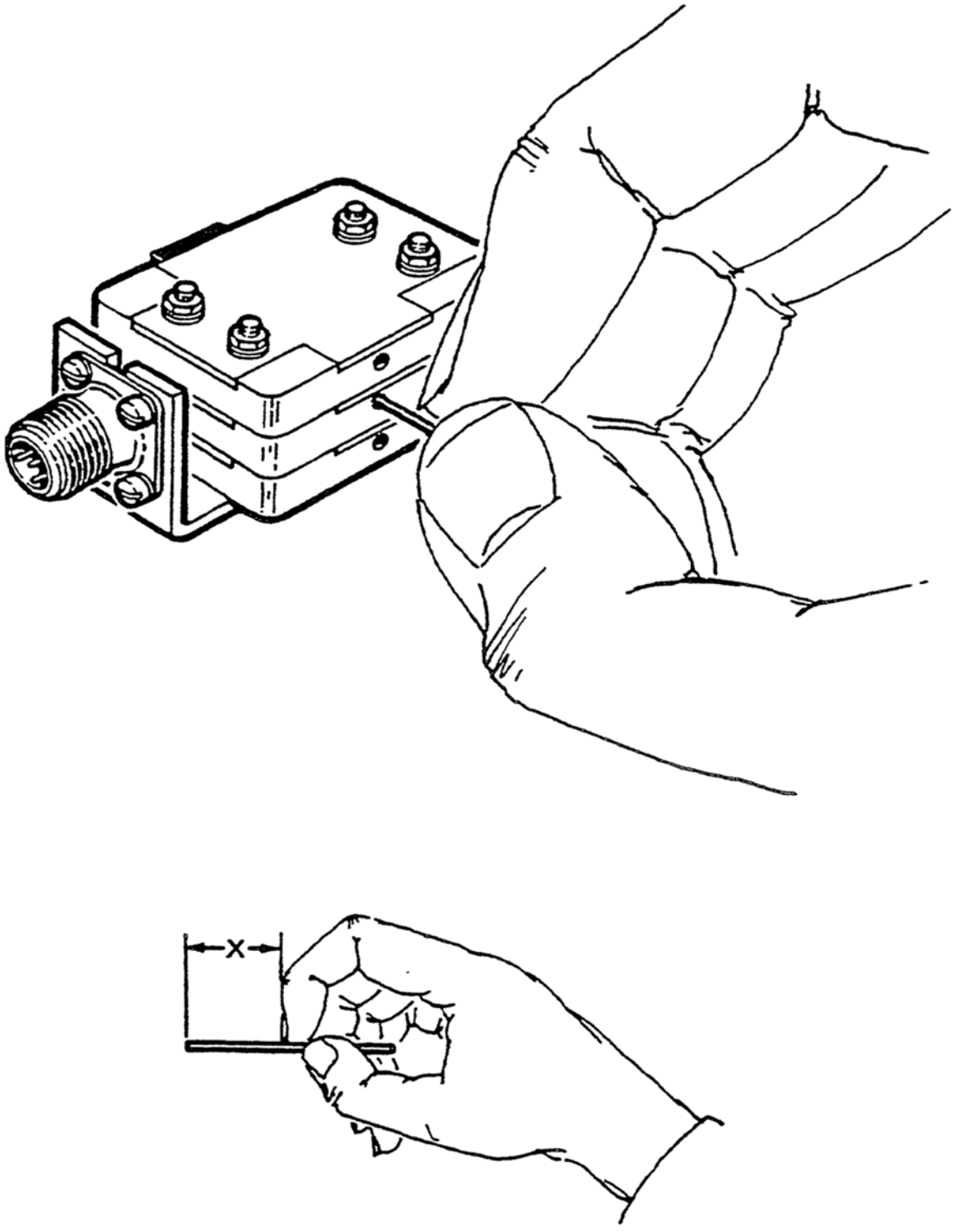
3. Inspection/Check

Refer to Figure 3 for details.

A. Check Brush Module Assembly, Brush Wear

1) Measure brush wear as shown in Figure 3. Replace brush retainer assembly when dimension X of any module is up to 23/64 inch maximum (Mod 6/1771).

Note During measurement allow only 1/16 inch of brush to protrude from module. This corresponds with normal brush extension when brush module assembly is installed on aircraft.



Brush Module Assembly – Checking Brush Wear
Figure 3